

Owen Chambers

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PROFILE

Applied AI engineer (MMath CS, University of Waterloo) building agentic development systems: LLM evaluation loops, MCP-backed git and Jira workflows, and AWS Lambda modernization for a Tier-1 Canadian bank. Claude Certified Architect Foundations. Prior contract work producing demonstration traces and code evaluations for LLM agents.

EXPERIENCE

Software Developer

BrainRidge Consulting

Nov 2024–Present

Spec-Driven Agentic SDLC Orchestrator

- Built an **LLM orchestration** platform that converted Jira requirements into pull requests via **code evaluation**, **test generation**, **spec ingestion**, and **git-delivery** agents using **AWS Bedrock**, **Anthropic API**, **GitHub MCP**, and **Atlassian MCP**.
- Crafted **prompts**, **skills**, **hooks**, **tools**, and **MCP server** integrations to improve consistency, traceability, and production safety.
- Implemented **DynamoDB**-backed state tracking and error tracking for **continual improvement**.
- Designed the review loop around typed envelopes, retry briefs, explicit halt conditions, and error categorization so long-running tasks could continue autonomously without losing context or quality control.

Autonomous AWS Migration

- Built a **Copilot CLI** workflow to modernize internal **AWS Lambda** services across 65+ applications at a Tier-1 Canadian bank, including **CDK** upgrades, runtime migration, and CVE/security remediation.
- Tracked the full **agentic lifecycle** from analyze → prepare → execute → test → review → handoff, iterating on prompts, tools, and guardrails based on actual agent runs and reviewer feedback.
- Reduced upgrade effort by roughly **75%** (~80 hours saved) while delivering the work through bank SDLC with zero production issues, using structured developer notes to halt unsafe changes early.
- Provided **training and documentation** for internal teams to adopt the workflow.

Agentic AWS Call Centre POC

- Worked on an agentic call centre POC using **AWS Connect**, **Lex** and the **Anthropic API** to answer customer calls, respond to questions, and escalate using **Claude Architect** best practices.
- Completed a full end to end workflow from customer call → agentic response → human handoff, including **prompt engineering** and **tool design**.

Machine Learning Contractor

Invisible Technologies

May 2024–Oct 2024

- Full-time contractor to produce human demonstration traces and code critiques for LLM agent evaluation.
- Completed multi-step web tasks in an instrumented environment so action sequences could be used as computer-use training/eval data.
- Ran model-generated code in a sandbox, reproduced failures, and submitted corrected implementations as labeled eval signal.

EDUCATION / RESEARCH

University of Waterloo

MMath in Computer Science - 3.7/4

Waterloo, Ontario, Canada

2021–2023

- Thesis: User-specific explanations of AI systems attuned to psychological profiles: a user study.

Brandon University

Honors B.Sc. Computer Science - 3.83/4.3

Brandon, Manitoba, Canada

2016–2021

PROJECTS

• PETR

- Personal **TypeScript** orchestration system for autonomous engineering work, modeled on a **Prepare / Execute / Test / Review** loop.
- Coordinates multi-step agent runs across repos with typed envelopes, retry briefs, explicit halt conditions, execution traces, and artifact logs for auditable unattended work.
- Emphasizes **tool-use discipline**, **validation gates**, and **recovery workflows** so iterative coding tasks can continue safely across long-running epics.

• Anchor Events

- Solo full-stack **Angular + Flask + PostgreSQL** application for sports trip discovery and itinerary planning.
- Built the frontend, API layer, database schema, and deployment flow for a live product at **anchorevents.ca**, with a focus on filtering, browsing, and planning event-driven travel experiences.
- Highlights **full-stack development**, **database design**, **user experience iteration**, and **production delivery** across the end-to-end product lifecycle.

SKILLS

- **Languages:** TypeScript, Python, HTML, CSS, SQL, Bash
- **Technologies:** AWS (Lambda, CDK, Bedrock, DynamoDB, Lex, Connect), Claude API / Agent SDK / MCP / Angular, Flask, SQLAlchemy, PostgreSQL

PUBLICATIONS

- [1] O. Chambers, R. Cohen, M. R. Grossman, L. Hebert, and E. Awad, “Mining user study data to judge the merit of a model for supporting user-specific explanations of ai systems”, *Computational Intelligence*, vol. 40, no. 6, e70015, 2024.
- [2] O. Chambers, R. Cohen, M. R. Grossman, and Q. Chen, “Creating a user model to support user-specific explanations of ai systems”, in *Adjunct Proceedings of the 30th ACM Conference on User Modeling, Adaptation and Personalization*, 2022, pp. 163–166.